An Analysis of Cross-validation statistics

Cross-validation statistics

Cross-validation statistics -- Part 1

Relative accuracy can be quantified as MSE (λ i) / MSE (λ R) {\displaystyle {\mbox{MSE}}(\lambda _{i})/{\mbox{MSE}}(\lambda _{R})} , so that the mean squared error of a candidate λ i {\displaystyle \lambda _{i}} is made relative to that of a user-specified λ R {\displaystyle \lambda _{R}} . The relative simplicity term measures the amount that λ i {\displaystyle \lambda _{i}} deviates from λ R {\displaystyle \lambda _{R}} relative to the maximum amount of deviation from λ R {\displaystyle \lambda _{R}} . Accordingly, relative simplicity can be specified as (λ i − λ R) 2 (λ max − λ R) 2 {\displaystyle {\frac {(\lambda _{i}-\lambda _{R})^{2}}{(\lambda _{\max }-\lambda _{R})^{2}}}} , where λ max {\displaystyle \lambda _{\max }} corresponds to the λ {\displaystyle \lambda } value with the highest permissible deviation from λ R {\displaystyle \lambda _{R}} . With γ ∈ [0 , 1] {\displaystyle \gamma \in [0,1]} , the user determines how high the influence of the reference parameter is relative to cross-validation. One can add relative simplicity terms for multiple configurations c = 1 , 2 , . . . , C {\displaystyle c=1,2,...,C} by specifying the loss function as

Cross-validation statistics -- Part 2

k-fold cross-validation In k-fold cross-validation, the original sample is randomly partitioned into k equal sized subsamples, often referred to as "folds". Of the k subsamples, a single subsample is retained as the validation data for testing the model, and the remaining k − 1 subsamples are used as training data. The cross-validation process is then repeated k times, with each of the k subsamples used exactly once as the validation data. The k results can then be averaged to produce a single estimation. The advantage of this method over repeated random sub-sampling (see below) is that all observations are used for both training and validation, and each observation is used for validation exactly once. 10-fold cross-validation is commonly used, but in general k remains an unfixed parameter. For example, setting k = 2 results in 2-fold cross-validation. In 2-fold cross-validation, the dataset is randomly shuffled into two sets d0 and d1, so that both sets are equal size (this is usually implemented by shuffling the data array and then splitting it in two). We then train on d0 and validate on d1, followed by training on d1 and validating on d0. When k = n (the number of observations), k-fold cross-validation is equivalent to leave-one-out cross-validation. In stratified k-fold cross-validation, the partitions are selected so that the mean response value is approximately equal in all the partitions. In the case of binary classification, this means that each partition contains roughly the same proportions of the two types of class labels. In repeated cross-validation the data is randomly split into k partitions several times. The performance of the model can thereby be averaged over several runs, but this is rarely desirable in practice. When many different statistical or machine learning models are being considered, greedy k-fold cross-validation can be used to quickly identify the most promising candidate models.

Cross-validation statistics -- Part 3

Using prior information When users apply cross-validation to select a good configuration λ {\displaystyle \lambda } , then they might want to balance the cross-validated choice with their own estimate of the configuration. In this way, they can attempt to counter the volatility of cross-validation when the sample size is small and include relevant information from previous research. In a forecasting combination exercise, for instance, cross-validation can be applied to estimate the weights that are assigned to each forecast. Since a simple equal-weighted forecast is difficult to beat, a penalty can be added for deviating from equal weights. Or, if cross-validation is applied to assign individual weights to observations, then one can penalize deviations from equal weights to avoid wasting potentially relevant information. Hoornweg (2018) shows how a tuning parameter γ {\displaystyle \gamma } can be defined so that a user can intuitively balance between the accuracy of cross-validation and the simplicity of sticking to a reference parameter λ R {\displaystyle \lambda _{R}} that is defined by the user. If λ i {\displaystyle \lambda _{i}} denotes the i t h {\displaystyle i^{th}} candidate configuration that might be selected, then the loss function that is to be minimized can be defined as

Cross-validation statistics -- Part 4

k*l-fold cross-validation This is a truly nested variant which contains an outer loop of k sets and an inner loop of l sets. The total data set is split into k sets. One by one, a set is selected as the (outer) test set and the k - 1 other sets are combined into the corresponding outer training set. This is repeated for each of the k sets. Each outer training set is further sub-divided into l sets. One by one, a set is selected as inner test (validation) set and the l - 1 other sets are combined into the corresponding inner training set. This is repeated for each of the l sets. The inner training sets are used to fit model parameters, while the inner test set is used as a validation set to provide an unbiased evaluation of the model fit. Typically, this is repeated for many different hyperparameters (or even different model types) and the validation set is used to determine the best hyperparameter set (and model type) for this inner training set. After this, a new model is fit on the entire outer training set, using the best set of hyperparameters from the inner cross-validation. The performance of this model is then evaluated using the outer test set.

Cross-validation statistics -- Part 5

Cross-validation, sometimes called rotation estimation or out-of-sample testing, is any of various similar model validation techniques for assessing how the results of a statistical analysis will generalize to an independent data set. Cross-validation includes resampling and sample splitting methods that use different portions of the data to test and train a model on different iterations. It is often used in settings where the goal is prediction, and one wants to estimate how accurately a predictive model will perform in practice. It can also be used to assess the quality of a fitted model and the stability of its parameters. In a prediction problem, a model is usually given a dataset of known data on which training is run (training dataset), and a dataset of unknown data (or first seen data) against which the model is tested (called the validation dataset or testing set). The goal of cross-validation is to test the model's ability to predict new data that was not used in estimating it, in order to flag problems like overfitting or selection bias and to give an insight on how the model will generalize to an independent dataset (i.e., an unknown dataset, for instance from a real problem). One round of cross-validation involves partitioning a sample of data into complementary subsets, performing the analysis on one subset (called the training set), and validating the analysis on the other subset (called the validation set or testing set). To reduce variability, in most methods multiple rounds of cross-validation are performed using different partitions, and the validation results are combined (e.g. averaged) over the rounds to give an estimate of the model's predictive performance. In summary, cross-validation combines (averages) measures of fitness in prediction to derive a more accurate estimate of model prediction performance.